MS S23.pdf

Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
	(d)	(i)		$K_c = \frac{[HOCl]^2}{[Cl_2O][H_2O]}$	1	1				
		(ii)	I	$[H_2O]$ at equilibrium = 0.29 mol dm^{-3} (1) $[Cl_2O]$ at equilibrium = 0.03 mol dm^{-3} (1) K_c = 5.56 (1) ecf possible from incorrect *p* expression	3		1 1	1	1 1 1	
			II	at a higher temperature K_c is larger so equilibrium has shifted to the right (1) reaction is endothermic (1) ecf possible from incorrect K_c value	2		1 1			
			III	Tl^+ and Fe^{2+} (1) SEP for HClO is more positive than for these half-cells so can oxidise the ions OR positive EMF values calculated for the reactions (1)	2		1 1			
					18	**2**	**12**	**4**	**7**	**0**